KARABO MOKOBANE

JUNIOR CYBERSECURITY / SOC ANALYST | IT SUPPORT & SECURITY OPERATIONS

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Cybersecurity portfolio: soc-security-homelab.vercel.app | GitHub: github.com/KaraboMokobane

PROFESSIONAL PROFILE

IT support and security operations professional with nearly three years of progressively responsible experience, including Service Desk Analyst responsibilities at the Cyril Ramaphosa Foundation. Brings hands-on exposure to Microsoft 365 administration, identity and access support, endpoint compliance, vulnerability awareness, network troubleshooting and incident handling. Building a focused Blue Team capability through a documented enterprise-style homelab using Wazuh, Security Onion, pfSense, OPNsense, Proxmox, Windows Server, Linux and segmented VLANs. Seeking a Junior SOC Analyst, Cybersecurity Analyst or Security Operations role where disciplined investigation, clear documentation and strong infrastructure fundamentals can support meaningful detection and response outcomes.

CYBERSECURITY FOCUS

- Security monitoring and log analysis: Wazuh, Security Onion, Zeek, Suricata, firewall logs, syslog and endpoint telemetry.
- Network defence: firewall policies, VLAN segmentation, TCP/IP, DNS, DHCP, VPNs, routing fundamentals and secure remote access.
- Identity and endpoint security: Microsoft Entra ID (Azure AD), Active Directory, Intune, MFA, Microsoft 365 and Sophos endpoint protection.
- Authorised lab security validation: Kali Linux, Nmap, Nessus, MITRE ATT&CK, MITRE Caldera, DVWA, WebGoat, Juice Shop and Metasploitable2.

PROFESSIONAL EXPERIENCE

Founder & Director

Tech Port | February 2025 - Present | Project-based IT services business

- Founded and operate a registered IT services business providing technical support, hardware support, websites, e-commerce solutions and practical cybersecurity assistance for small-business clients.
- Deliver WordPress, Elementor and WooCommerce implementations, including PayFast payment setup, SSL/TLS configuration, web application firewall settings and ongoing maintenance support.
- Manage client discovery, quotations, support arrangements, technical documentation and service delivery while applying a security-conscious approach to online services and customer systems.

Service Desk Analyst

Cyril Ramaphosa Foundation | June 2025 - May 2026

- Resolved escalated hardware, software, Microsoft 365 and connectivity incidents; captured clear incident records, performed structured troubleshooting and coordinated escalation within service-level expectations.
- Administered Microsoft 365 services including Exchange Online, SharePoint, OneDrive and Teams; supported licensing, user accounts, permissions, onboarding and offboarding activities.
- Supported identity and endpoint controls through Microsoft Intune enrolment, device compliance checks, multi-factor authentication resets, Sophos endpoint protection and VPN access troubleshooting.
- Contributed to cloud migration and infrastructure changes with Smart Technology Centre, supporting adoption of Microsoft 365, SharePoint Online and OneDrive services.
- Supported RAM upgrades, office moves, asset checks, server backup and disposal activity; maintained documentation and communicated technical progress to users and stakeholders.

PROFESSIONAL EXPERIENCE CONTINUED

IT Support Intern

Cyril Ramaphosa Foundation | July 2023 - May 2025

- Provided frontline support for Windows and macOS devices, printers, peripherals, productivity applications and remote users, diagnosing hardware, software and network connectivity faults.
- Configured workstations and user environments; supported Active Directory accounts, group and permission administration, Microsoft 365 services, VPN access and Remote Desktop Protocol connectivity.
- Assisted users with Exchange Online, SharePoint, OneDrive and Teams, helping protect business continuity through dependable access and user-focused technical guidance.
- Participated in endpoint security and cloud migration initiatives, including laptop and OneDrive compliance checks, asset reviews and user adoption support.
- Produced and improved technical documentation and process maps, strengthening repeatable support practices and clearer handover to service partners.

IT Training Programme / Technical Mentorship

Smart Technology Centre | 2023 - 2024 | Professional development

- Participated in monthly, scenario-based technical training covering IT support, systems administration, Microsoft 365 administration, endpoint support and infrastructure troubleshooting.
- Gained practical exposure to diagnosing hardware, software and network issues across Windows and macOS environments, with emphasis on DNS, DHCP, VPN and user provisioning fundamentals.
- Contributed to an organisational migration from on-premises services to Microsoft 365, SharePoint Online and OneDrive alongside experienced infrastructure professionals.

CYBERSECURITY PROJECTS

Enterprise Cybersecurity Homelab

Independent technical portfolio project | 2026 - Present

- Designed and operate a two-node Proxmox VE virtualisation lab with Windows and Linux workloads, pfSense and OPNsense firewalls, Docker services and separated user, security and server networks.
- Implemented VLAN segmentation and firewall policy controls across VLAN 10, VLAN 20 and VLAN 30; configured Windows Server DHCP for a dedicated lab network and validated host connectivity.
- Deployed Wazuh with Linux endpoint agents; reviewed agent health, event visibility and ingestion issues while documenting remediation steps for connectivity, storage and log collection problems.
- Configured pfSense remote syslog and firewall-rule logging to support central visibility; investigated the difference between firewall evidence, endpoint telemetry and SIEM search results.
- Installed Security Onion with Zeek and Suricata to build network-monitoring skills; observed connection metadata and documented detection gaps that require rule tuning or sensor verification.
- Maintain portfolio evidence including topology diagrams, screenshots, incident notes, lessons learned and lab change records at soc-security-homelab.vercel.app.

SELECTED SECURITY INVESTIGATIONS AND LEARNING

Incident 001 - Network Discovery and Reconnaissance

Authorised lab exercise | 2026 | MITRE ATT&CK T1046

- Performed controlled Nmap discovery from Kali Linux against an Ubuntu server hosting intentionally vulnerable Docker applications in a segregated lab environment.
- Captured and reviewed evidence across pfSense firewall logs, Security Onion/Zeek network data and Wazuh endpoint monitoring to compare what each control observed.
- Documented exposed services, network paths, telemetry limitations and follow-up work, including the need to validate Suricata alerting and tune detection coverage.

Incident 002 - Web Application Attack Validation

Authorised lab exercise | 2026 | OWASP and application-security learning

- Used deliberately vulnerable applications, including DVWA and WebGoat, to document reflected XSS, stored XSS, insecure direct object reference (IDOR) behaviour and password-reset host-header poisoning in a controlled environment.
- Captured requests, responses and browser evidence to practise clear security reporting, reproduce findings and relate observations to broken access control, input handling and authentication-security concepts.
- Used the exercise to strengthen understanding of how web-application weaknesses can lead to unauthorised data access or malicious client-side execution when controls are absent.

Local AI Security Lab

Independent technical portfolio project | 2026 - Present

- Hosted Ollama and Open WebUI on an Ubuntu server in the lab and enrolled the host with Wazuh to establish baseline monitoring around an internally hosted AI workload.
- Exploring agentic-AI security concepts, including prompt injection, tool poisoning and workflow risks, following the AWS Summit 2026 session: Red Team vs Blue Team in the AI Era.

TECHNICAL SKILLS

Security operations and monitoring: Wazuh SIEM, endpoint agents, syslog, pfSense firewall logs, Security Onion, Zeek, Suricata, event review, alert triage foundations, incident documentation and MITRE ATT&CK mapping.

Security testing in authorised labs: Kali Linux, Nmap, Nessus, MITRE Caldera, vulnerability assessment fundamentals, web security testing with DVWA/WebGoat/Juice Shop, OWASP Top 10 concepts and Metasploitable2.

Networks and infrastructure: pfSense, OPNsense, Proxmox VE, Docker, Portainer, LXC, VLANs, TCP/IP, IPv4/IPv6, DNS, DHCP, VPN, Tailscale, Windows Server, Windows 10/11, Ubuntu, Linux Mint and macOS.

Identity, endpoint and Microsoft cloud: Microsoft 365, Exchange Online, SharePoint, OneDrive, Teams, Microsoft Intune, Active Directory, Microsoft Entra ID (Azure AD), Group Policy, MFA, CodeTwo and Sophos endpoint protection.

Automation, development and data: PowerShell, Python, SQL, C#, JavaScript, HTML/CSS, .NET, ASP.NET, Blazor, Entity Framework Core, React, Tailwind CSS, SQL Server, Power BI, Git/GitHub and VS Code.

Documentation and service delivery: ITSM ticketing, incident tracking, escalation, SLA awareness, technical writing, process mapping, draw.io, stakeholder communication and customer support.

EDUCATION AND CERTIFICATIONS

Bachelor of Computer Information Systems

University of the Free State | 2018 - 2022

- Relevant academic foundations: C# and Python programming, computer networks, database management (SQL), computer hardware, software engineering, systems analysis and design, and digital marketing.

Oracle Cloud Infrastructure Foundations Associate

Oracle University | 2025

- Cloud infrastructure, networking, security fundamentals, identity and access management, storage, monitoring and cloud service administration.

Oracle Cloud Infrastructure AI Foundations Associate

Oracle University | 2025

- AI and machine-learning concepts, OCI AI services, data concepts, responsible AI and business use cases.

Essentials of Network Addressing and Services

British Columbia Institute of Technology | 2023

- IPv4/IPv6 addressing, DHCP, DNS, routing, core protocols and network-service fundamentals.

Essentials of Network Terminology and Protocols

British Columbia Institute of Technology | 2023

- Network architecture, protocols, OSI model, media types, network simulations and basic network design.

Business Analysis for Innovation and Transformation

NoviaOne Business School | 2025

- Business process improvement, design thinking, data-informed decision-making, business transformation and alignment of IT solutions to operational needs.

CURRENT PROFESSIONAL DEVELOPMENT

- Cybersecurity Operations coursework, British Columbia Institute of Technology - in progress.
- CompTIA Security+ (SY0-701) self-study - building formal knowledge of security operations, identity, governance, incident response and network security.
- Google IT Support coursework - in progress.
- Transactional Banking Programme, NoviaOne (2025 - 2026) - completed modules in transactional banking, product and solution design, data and analytics, financial analysis, client context and professional skills.
- AWS Summit 2026 - attended Red Team vs Blue Team in the AI Era, focused on AI-agent security risks and defensive thinking.

ADDITIONAL INFORMATION

Work eligibility: South Africa. References available on request. Portfolio projects are completed only in personally owned, deliberately vulnerable or otherwise authorised lab environments.